

Complexity Bounds for Dirichlet Process Slice Samplers

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Joint work with:
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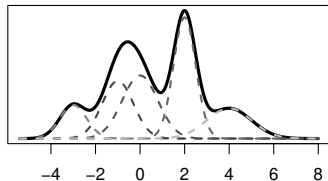
DP-based models

$$\begin{array}{ccc}
 Y \mid \eta, \mathbf{z}_n \sim \underbrace{\mathcal{L}(\mathbf{z}_n, \eta)}_{\substack{\text{likelihood} \\ \text{depends on} \\ \mathbf{z}_n = (z_1, \dots, z_n)}} & z_i \mid G \stackrel{iid}{\sim} \underbrace{G(dx) \stackrel{a.s.}{=} \sum_{k=1}^{\infty} \pi_k \delta_{\phi_k}(dx)}_{\substack{\text{discrete random} \\ \text{probability measure}}} & \underbrace{G \sim \text{DP}(\alpha, P_0)}_{\text{prior}}
 \end{array}$$

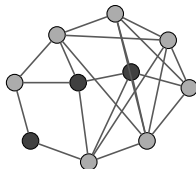
$$G \sim \text{DP}(\alpha, P_0) \implies \phi_k \stackrel{iid}{\sim} P_0, \quad V_k \stackrel{iid}{\sim} \text{Beta}(1, \alpha) \quad \pi_k = V_k \prod_{\ell < k} (1 - V_\ell)$$

$$\implies \rho_n \text{ partition of } [n] = \{1, \dots, n\} \text{ defined by } i \sim j \text{ if and only if } z_i = z_j$$

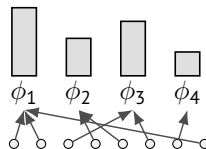
Mixture models



Stochastic block models



Species sampling models



MCMC sampling in DP-based models

$$Y \mid \eta, \mathbf{z}_n \sim \underbrace{\mathcal{L}(\mathbf{z}_n, \eta)}_{\substack{\text{likelihood} \\ \text{depends on} \\ \mathbf{z}_n = (z_1, \dots, z_n)}} \quad z_i \mid G \stackrel{iid}{\sim} \underbrace{G(dx) \stackrel{a.s.}{=} \sum_{k=1}^{\infty} \pi_k \delta_{\phi_k}(dx)}_{\substack{\text{discrete random} \\ \text{probability measure}}} \quad \underbrace{G \sim \text{DP}(\alpha, P_0)}_{\text{prior}}$$

Marginal algorithms

(Neal, 2000)

$$\text{pr}(z_i = z | \text{rest}) \propto \begin{cases} n_h(\mathbf{z}_n^{-(i)}) \times \mathcal{L}(\mathbf{z}_n, \eta) & \text{if } z \in \mathbf{z}_n^{-(i)} \\ \alpha \times \int \mathcal{L}(\mathbf{z}_n, \eta) P_0(dz) & \text{if } z = \text{new} \end{cases}$$

- ✓ Target the posterior
- ✗ Force one-at-a-time updates
- ✗ Bookkeeping

Blocked Gibbs samplers

(Ishwaran and James, 2001)

$$\text{pr}(z_i = z | \text{rest}) \propto \pi_{k(z)} \times \mathcal{L}(\mathbf{z}_n, \eta)$$

- ✗ Target an approximation of the posterior
- ✓ Allow joint updates
- ✓ Easy/No bookkeeping

Non-vanishing truncation bias of blocked Gibbs samplers

Exponential accuracy of truncated blocked Gibbs samplers (Ishwaran and James, 2002)

Let L be the truncation level, i.e., G is approximated by $\sum_{\ell=1}^L \pi_{\ell} \delta_{\phi_{\ell}}$ and ρ_n the partition induced by ties in z_i in a Normal mixture model

$$\int_{\mathbb{R}^n} \sum_{\rho_n} \left| \underbrace{p_L(\rho_n \mid \mathbf{y}_{1:n})}_{\text{target of the algorithm}} - \underbrace{p_{\infty}(\rho_n \mid \mathbf{y}_{1:n})}_{\text{posterior of the partition}} \right| \underbrace{m_{\infty}(\mathbf{y}_{1:n})}_{\text{prior predictive}} d\mathbf{y}_{1:n} = O(4 n e^{-(L-1)/\alpha}).$$

Extensions in Ishwaran and Zarepour (2002); Campbell et al. (2019); Li and Campbell (2021).

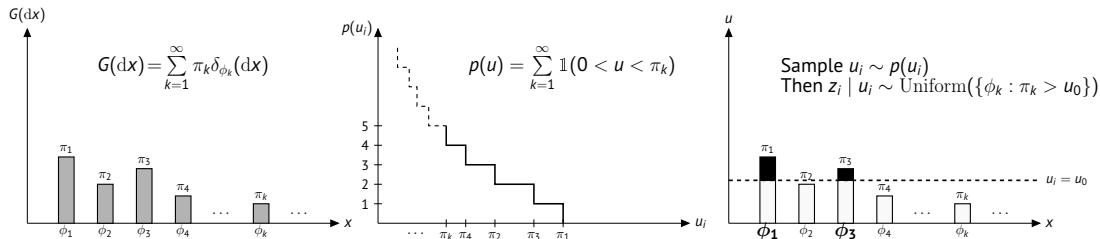
Two main limitations of the result:

✗ It holds **under the generative model**

✗ Trivially: unless $L = n$, $p_L(\rho_n \mid \mathbf{y}_{1:n}) = 0$ on all partitions with more than L clusters.

Slice sampling (Neal, 2003; Walker, 2007)

$$z_i \sim G(dz) \quad \rightarrow \quad (u_i, z_i) \sim \mathbb{1}(0 < u < G(dz))$$



- ✓ Target the posterior
- ✓ Allow joint updates
- ✓ Easy/No bookkeeping

Slice sampling (Neal, 2003; Walker, 2007; Ge et al., 2015)

Posterior slice sampling in DP-based models

1. Sample G , i.e., $(\pi_1, \pi_2, \dots)$ and $(\phi_1, \phi_2, \dots)$
2. Sample u_i from $p(u_i | z_i = \phi_k) = \text{Uniform}(0, \pi_k) \quad \forall i$
3. Sample z_i from $p(z_i | u_i = u_0) = \text{Discrete}(\phi_k : \pi_k > u_i, \text{ with } w_k \propto \text{Lik}(Y; z_i = \phi_k)) \quad \forall i$

How many $(\pi_h, \phi_k)'s$ do we need to sample at step 1?

→ Step 2. requires weights for all currently allocated components.

$$\underbrace{(\pi_1, \dots, \pi_H, \pi^*)}_{H=\text{number of clusters}} \sim \text{Dirichlet}(n_1, \dots, n_H, \alpha) \quad (\text{Ge et al., 2015})$$

→ Step 3. requires determination of the slice set $A_i = \{\phi_k : \pi_k > u_i\}$, let $u_{\min} = \min_i u_i$

$$\forall i, A_i \subseteq \left\{ \phi_1, \dots, \phi_K : \sum_{h=K+1}^{\infty} \pi_h < u_{\min} \right\} \quad (\text{Walker, 2007})$$

Dynamic truncation level K

$$K = \min \left\{ k \geq H : \sum_{h=k+1}^{\infty} \pi_h < u_{\min} \right\}$$

✓ At each iteration, K components are enough and K is finite.
✗ Time complexity is driven by at most $K \times n$ likelihood evaluations.

About K

$$K = \min \left\{ k \geq H : \sum_{h=k+1}^{\infty} \pi_h < u_{\min} \right\}$$

✗ If $u_{\min} \approx 0$, K can be arbitrarily large.

$\mathbb{P}_{\pi}(\exists s \leq T : C_s > \bar{C}) \xrightarrow{T \rightarrow \infty} 1$

stationary distribution (points to $\mathbb{P}_{\pi}$)

any finite threshold (points to $\bar{C}$)

*computational cost of iteration s
(in some unit of measure)* (points to C_s)

→ **No deterministic bound on per-iteration time exists.**

Informally: To assess actual scalability, we need to quantify the “probability of exceeding a threshold” and how this varies with n .

About K (from now K_n)

A related quantity: $K(x) = \min \left\{ k \geq 1 : \sum_{h=k+1}^{\infty} \underbrace{\pi_h}_{\text{from the prior}} < x \right\} \quad x \in (0, 1)$

From Muliere and Tardella (1998)

$$K \mid x \sim 1 + \text{Poisson}(-\alpha \log x)$$

But at **each MCMC iteration** (in the improved slice sampler of Ge et al., 2015)

$$K_n = \min \left\{ k \geq H_n : \sum_{h=k+1}^{\infty} \pi_h < u_{\min} \right\}$$

where:

H_n is the number of clusters of the partition ρ_n visited by the chain

π_h 's depend on the partition ρ_n visited by the chain

$u_{\min}$ depends on the $(\pi_h)_{h \geq 1}$ and on the partition ρ_n visited by the chain

ρ_n depends on the specific data

Survival probability of $u_{\min}$

Proposition: Merging two clusters increases the survival probability of $u_{\min}$

If $\rho_n^{(r \oplus s)}$ is the partition obtained from ρ_n by merging two (distinct) clusters r and s into a single cluster and leaving the other clusters unchanged, then, for any $x \in (0, 1)$

$$\Pr(u_{\min} > x \mid \rho_n^{(r \oplus s)}) \geq \Pr(u_{\min} > x \mid \rho_n).$$

Corollary: Singleton partition yields the lowest survival probability of $u_{\min}$

Let ρ_n^{sing} denote the singleton partition of $[n]$, i.e., the partition with n clusters of size 1. Then, for every $x \in (0, 1)$ and every partition ρ_n ,

$$\Pr(u_{\min} > x \mid \rho_n) \geq \Pr(u_{\min} > x \mid \rho_n^{\text{sing}})$$

Main result

Theorem: High-probability bound on dynamic truncation level (Franzolini and Gaffi, 2026)

Let K_n and H_n be respectively the truncation level and the number of occupied clusters at a given iteration when run for n input data.

Then, for every $\delta \in (0, 1)$ and any $n \geq 2$

$$\Pr(K_n - H_n \leq C_\delta \log n \mid \rho_n) \geq 1 - \delta, \quad \forall \rho_n$$

with $C_\delta = B_\alpha^{(1)} + B_\alpha^{(2)} \log(1/\delta)$, and $B_\alpha^{(1)}, B_\alpha^{(2)} \asymp \alpha$.

In particular, as $n \rightarrow \infty$, $K_n - H_n = O_{\mathbb{P}}(\log n)$ and, in the worst-case scenario of $H_n = O(n)$, $K_n = O_{\mathbb{P}}(n)$.

“In the worst case, the slice sampler requires $O(n)$ active components with high probability.”

Proof sketch

$$K_n = \min \left\{ k \geq H_n : \sum_{h=k+1}^{\infty} \pi_h < u_{\min} \right\} = \min \left\{ k \geq H_n : \pi^{\star} \times \prod_{i=H_n+1}^k (1 - V_i) < u_{\min} \right\}.$$

Proof sketch

$$K_n = \min \left\{ k \geq H_n : \sum_{h=k+1}^{\infty} \pi_h < u_{\min} \right\} = \min \left\{ k \geq H_n : \pi^* \times \prod_{i=H_n+1}^k (1 - V_i) < u_{\min} \right\}.$$

■ Define $K_n^0(x) = \min \left\{ k > H_n : \prod_{i=H_n+1}^k (1 - V_i) < x \right\}$

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■ $\Pr(K_n - H_n > C_\delta \log n \mid \rho_n) \leq \boxed{\Pr(K_n^0(x) > H_n + C_\delta \log n \mid \rho_n)} + \boxed{\Pr(K_n > K_n^0(x) \mid \rho_n)}$

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■ Show that $\boxed{\Pr(K_n^0(x) \geq 1 + H_n + 3\alpha \log(1/x) + \log(2/\delta) \mid \rho_n) \leq \frac{\delta}{2}} \quad (1)$

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■ Show that $\boxed{\Pr(K_n > K_n^0(x) \mid \rho_n) \leq \Pr(u_{\min} \leq x \mid \rho_n) \leq n(\alpha + n - 1)x[1 + \log(1/x)]} \quad (2)$

Proof sketch

$$K_n = \min \left\{ k \geq H_n : \sum_{h=k+1}^{\infty} \pi_h < u_{\min} \right\} = \min \left\{ k \geq H_n : \pi^* \times \prod_{i=H_n+1}^k (1 - V_i) < u_{\min} \right\}.$$

■ Define $K_n^0(x) = \min \left\{ k > H_n : \prod_{i=H_n+1}^k (1 - V_i) < x \right\}$

■ $\Pr(K_n - H_n > C_\delta \log n \mid \rho_n) \leq \boxed{\Pr(K_n^0(x) > H_n + C_\delta \log n \mid \rho_n)} + \boxed{\Pr(K_n > K_n^0(x) \mid \rho_n)}$

■ Show that $\boxed{\Pr(K_n^0(x) \geq 1 + H_n + 3\alpha \log(1/x) + \log(2/\delta) \mid \rho_n) \leq \frac{\delta}{2}} \quad (1)$

■ Show that $\boxed{\Pr(K_n > K_n^0(x) \mid \rho_n) \leq \Pr(u_{\min} \leq x \mid \rho_n) \leq n(\alpha + n - 1)x[1 + \log(1/x)]} \quad (2)$

■ Choose $x = x(n, \delta) \in (0, 1)$ such that, for any $n \geq 2$, the lower-bound in the probability in (1) is upper-bounded by $H_n + C_\delta \log n$, and the r.h.s. of (2) is upper-bounded by $\delta/2$.

A few additional results

Corollary: exponential tails of the slice overhead

Let (K_n, H_n) be defined as in the main Theorem.

There exist constants $B_\alpha^{(1)}, B_\alpha^{(2)} > 0$ such that for all $n \geq 2$ and all $t \geq 0$,

$$\Pr \left(\frac{K_n - H_n}{\log n} > B_\alpha^{(1)} + B_\alpha^{(2)} t \mid \rho_n \right) \leq e^{-t}, \quad \forall \rho_n.$$

Corollary: Almost-sure control of super-logarithmic slice overheads with infinite data coupling

Let (K_n, H_n) be defined as in the main Theorem.

There exists a constant $D_\alpha > 0$ such that

$$\Pr \left(\limsup_{n \rightarrow \infty} \left\{ K_n - H_n > D_\alpha \log \frac{1}{\delta_n} \log n \right\} \right) = 0$$

for any summable sequence $(\delta_n)_{n \geq 1} \subset (0, 1/2)$.

A few additional results

Proposition: tightness of the logarithmic slice-overhead bound

Let (K_n, H_n) be defined as in the main Theorem.

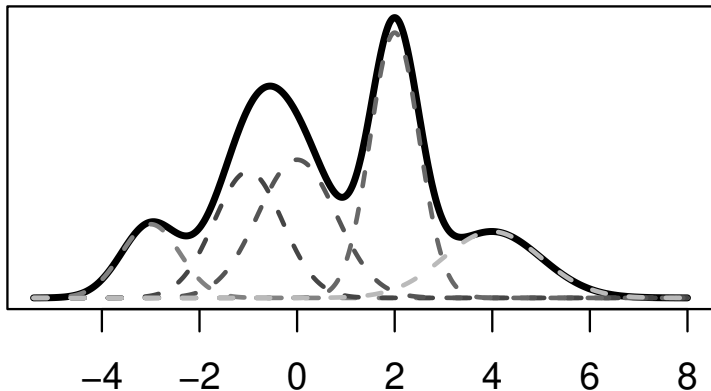
There exist constants $c_\alpha, \eta_\alpha > 0$ such that for all sufficiently large n ,

$$\Pr \left(K_n - H_n \geq c_\alpha \log n \mid \rho_n^{\text{sing}} \right) \geq \eta_\alpha.$$

Consequently, for any deterministic sequence $r_n = o(\log n)$ and any $M > 0$,

$$\liminf_{n \rightarrow \infty} \sup_{\rho_n} \Pr (K_n - H_n > Mr_n \mid \rho_n) > 0.$$

Practical consequences for mixture models



MCMC alternatives comparison

for $i \in \{1, \dots, n\}$ $Y_i \mid z_i \stackrel{\text{ind}}{\sim} \mathcal{N}(z_i, \sigma^2), \quad z_i \mid G \stackrel{\text{iid}}{\sim} G, \quad G \sim \text{DP}(\alpha, P_0),$

	CRP	BGS-L	BGS-n	Slice
Scalability by posterior cluster growth				
$H_n = O(n)$	$O(n^2)$	$\Theta(n)$	$\Theta(n^2)$	$O_{\mathbb{P}}(n^2)$
$H_n = O(\log n)$	$O(n \log n)$	$\Theta(n)$	$\Theta(n^2)$	$O_{\mathbb{P}}(n \log n)$
$H_n = O(1)$	$O(n)$	$\Theta(n)$	$\Theta(n^2)$	$O_{\mathbb{P}}(n \log n)$
Exact posterior partition target	✓	✗	✗	✓
No hard-threshold bias	✓	✗	✓	✓
No bookkeeping	✗	✓	✓	✓
Joint Updates	✗	✓	✓	✓

Numerical study

Competing algorithms:

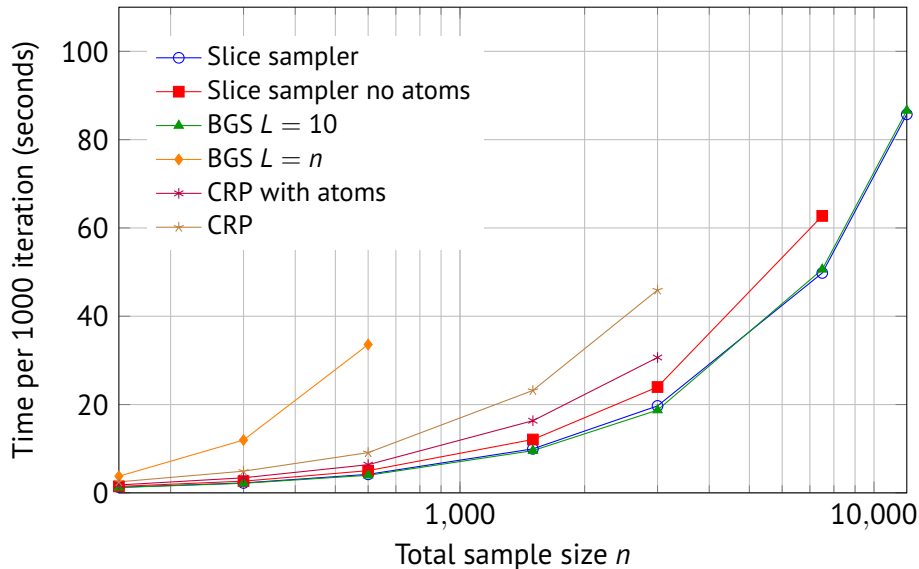
- ★ Slice sampler
- ★ Slice sampler with no atoms
- ★ BGS $L = 10$
- ★ BGS $L = n$
- ★ Marginal CRP
- ★ Marginal CRP with atoms

Simulated data scenario: Three clusters of equal size

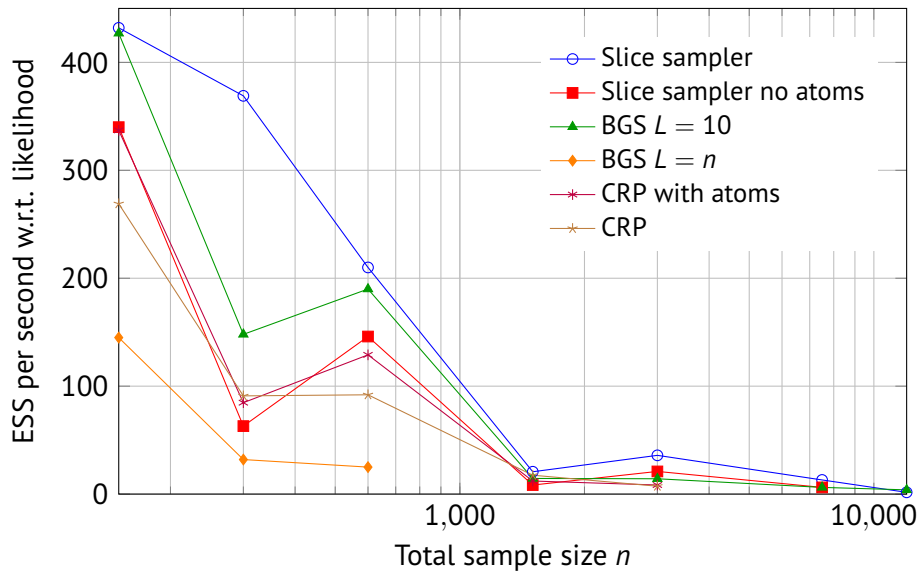
$$y_i \sim \mathcal{N}(\phi_{z_i}, \tau^{-1})$$

with true cluster means $\phi = (-3, 0, 3)$ and precision $\tau = 1$

Wall-clock time



Mixing



Numerical study

Competing algorithms:

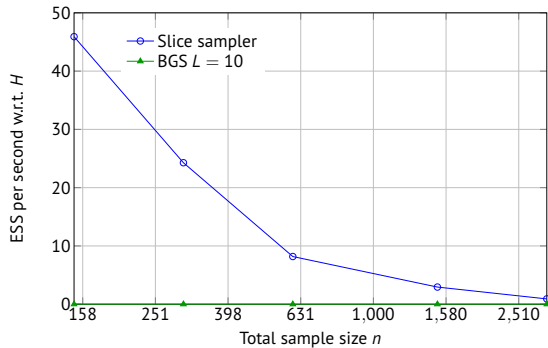
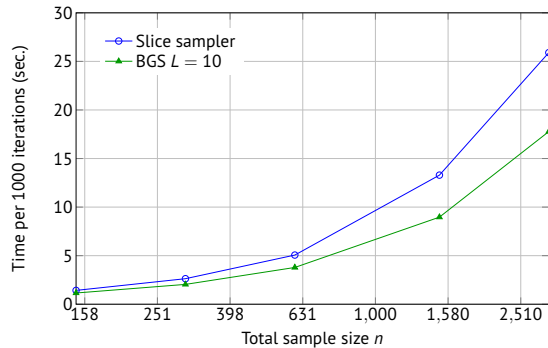
- ★ Slice sampler
- ★ BGS $L = 10$

Simulated data scenario: Perturbed Zipf

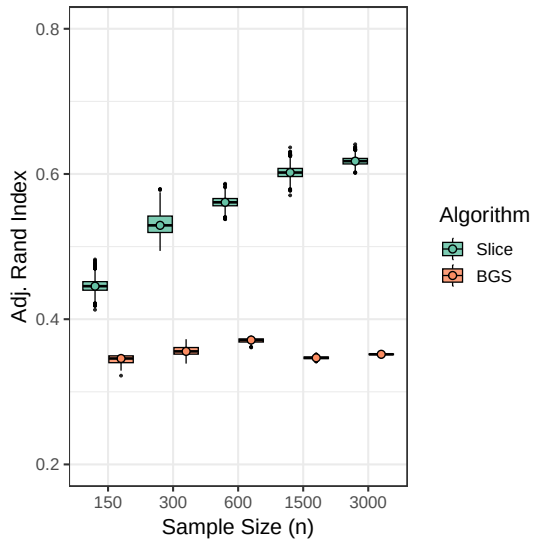
$$y_i \sim \mathcal{N}(\phi_{z_i}, \tau^{-1})$$

$$\phi_{z_i} = 3 \times z_i \quad p(z_i) \propto z_i^{-0.5} \quad \text{for } z_i \in \{1, \dots, 50\}$$

Wall-clock time and mixing



Inferential performance



Conclusions and future directions

Take-home message

- Slice samplers for DP-based models have **unbounded** per-iteration cost:

$$K_n = \min \left\{ k \geq H_n : \sum_{h=k+1}^{\infty} \pi_h < u_{\min} \right\}.$$

- Nevertheless, for **any dataset**, the slice overhead is **small with high probability**:

$$K_n - H_n = O_{\mathbb{P}}(\log n), \quad \text{uniformly over all visited partitions } \rho_n.$$

Future directions

- Extend the bounds to **adaptive DP models**, where the concentration parameter α is learned from the data.
- Study **richer BNP priors**: Pitman–Yor processes (Canale et al., 2022) and general species sampling processes (Mena et al., 2025).
- Develop guarantees for **structured models**: hierarchical DPs, sticky constructions, and dependent or temporally evolving partitions.

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